IDENTIFYING INFORMATION:

NAME: Skjellum, Anthony

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-5252-6600

POSITION TITLE: Professor

PRIMARY ORGANIZATION AND LOCATION: Tennessee Technological University, Computer Science Dept., Cookeville, Tennessee, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
California Institute of Technology , Pasadena, California, United States	PHD	09/1985 - 06/1990	Chemical Engineering and Computer Science (Minor)
California Institute of Technology , Pasadena, California, United States	MS	09/1984 - 06/1985	Chemical Engineering
California Institute of Technology , Pasadena, California, United States	BS	09/1980 - 06/1984	Physics

Appointments and Positions

2023 - present	Professor, Tennessee Technological University, Computer Science Dept., Cookeville, Tennessee, United States
2025 - present	Director, ASCEND Center, Tennessee Technological University, Cookeville, Tennessee, United States
2017 - 2023	Chair of Excellence, Professor of Computer Science, Director, SimCenter, UTC, Chattanooga, Tennessee, United States
2016 - 2017	McCrary Eminent Scholar Endowed Chair; Director , Charles D. McCrary Institute, Auburn, Alabama, United States
2014 - 2017	Professor of Computer Science and Software Engineering, Director, Auburn Lead Scientist for Cyber Research , Auburn Cyber Research Center (AUCRC), Auburn, Alabama, United States
2014 - 2016	Corporation Cyber Security and Information Assurance Endowed Professor, COLSA, Auburn, Alabama, United States
2009 - present	Co-Founder and CTO, RunTime Computing Solutions, LLC, Chattanooga, Tennessee, United States
2007 - 2009	Chief Architect, Verari Systems, Inc, Birmingham, Alabama, United States
2004 - 2007	Chief Software Architect, Verari Systems Software, Inc., Birmingham, Alabama, United States
2003 - 2014	Professor and Chair, University of Alabama at Birmingham (UAB), Dept. of Computer and Information Sciences, Birmingham, Alabama, United States

2002 - 2004	Chief Technology Officer, MPI Software Technology, Inc, Birmingham, Alabama, United States
2000 - 2003	Director, High Performance Computing Laboratory at the Dept. of Computer Science, MSU, Starkville, Mississippi, United States
1997 - 2003	Associate Professor (tenured), Dept. of Computer Science, Mississippi State University, Starkville, Mississippi, United States
1997 - 2000	Director, High Performance Computing Laboratory at the Engineering Research Center for Computational Field Simulation (ERC), Mississippi State University, Starkville, Mississippi, United States
1996 - 2002	Co-Founder I President, MPI Software Technology, Inc. , Starkville, Mississippi, United States
1996 - 1997	Associate Professor , Dept. of Computer Science, Mississippi State U, Starkville, Mississippi, United States
1993 - 1996	Assistant Professor , Dept. of Computer Science, Mississippi State U, Starkville, Mississippi, United States
1990 - 1993	Computer Scientist, Department of Energy, Lawrence Livermore National Laboratory (LLNL), Starkville, Mississippi, United States

Products

1. Gropp W, Lusk E, Doss N, Skjellum A. A High-Performance, Portable Implementation of the MPI Message-Passing Interface Standard,. Parallel Computing. 1996 September; 22(6):798-828.
2. Dimitrov R, Skjellum: A. Software Architecture and Performance Comparison of MPI/Pro and MPICH. International Conference on Computational Science; 2003 June. Other: pages 307-315
3. Dosanjh MGF, Worley A, Schafer D, Soundarajan P, Ghafoor S, Skjellum A, Bangalore PV, Grant RE. Implementation and evaluation of MPI 4.0 partitioned communication libraries. Parallel Computing. 2021 December 01; 108. Available from: http://www.mpi-forum.org/docs/docs.html DOI: 10.1016/j.parco.2021.102827
4. Skjellum A, Rüfenacht M, Sultana N, Schafer D, Laguna I, Mohror . ExaMPI: A new middleware architecture and implementation to accelerate innovation with the Message Passing Interface. CARLA. 2019.
5. Holmes DJ, Morgan B, Skjellum A, Bangalore PV, Sridharan . Planning for performance: Enhancing achievable performance for MPI through persistent collective operations. Parallel Computing. 2019; 81:32-57.
6. Schafer D, Ghafoor , Holmes J, Rüfenacht , Skjellum . Extending the Message Passing Interface (MPI) with User-Level Schedules. 26th IEEE Int. Conf. on High Performance Computing, Data, and Analytics; 2019.
7. Grant RE, Dosanjh G.F., Levenhagen J, Brightwell R, Skjellum . Finepoints: Partitioned Multithreaded MPI Communication. ISC. 2019; :330-350.
8. Jocksch A, Avans C, Shipley R, Skjellum A. A compilation-based approach to performant reduction and redistribution collective communication algorithms. The International Journal of High Performance Computing Applications. 2026; 40(2):219-239.

9. Abouelmagd A, Boehme D, Brink S, Burmark J, McKinsey M, Skjellum A, Pearce O. GPU Partitioning, Power, and Performance of the AMD MI300A. HPC Asia 2026; 2026 January; , Japan. c2026.
10. Bridges P, Schafer D, Lange J, White III J, Skjellum A, Suggs E, Hines T, Bangalore P, Dosanjh M, Schonbein W. Co-Design and Evaluation of a CPU-Free MPI GPU Communication Abstraction and Implementation. arXiv. 2026.

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Skjellum, Anthony in SciENcv on 2026-04-19 20:17:13